

Comparison Test for Positive Series

Topic.

Numerical series, positive sequences, comparison test.

Statement.

Let

$$a_n > 0, \quad b_n > 0,$$

and suppose that

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}, \quad \forall n \geq n_0.$$

Prove that

$$\sum_{n=1}^{\infty} b_n \text{ converges} \implies \sum_{n=1}^{\infty} a_n \text{ converges.}$$

Solution.

Since

$$a_n > 0, \quad b_n > 0,$$

we can transform the inequality:

$$\frac{a_{n+1}}{a_n} \leq \frac{b_{n+1}}{b_n}, \quad \forall n \geq n_0.$$

Then

$$\frac{a_{n+1}}{b_{n+1}} \leq \frac{a_n}{b_n}, \quad \forall n \geq n_0.$$

Hence the sequence

$$\left(\frac{a_n}{b_n} \right)$$

is non-increasing for

$$n \geq n_0.$$

Therefore,

$$\frac{a_n}{b_n} \leq \frac{a_{n_0}}{b_{n_0}}, \quad \forall n \geq n_0.$$

Put

$$C = \frac{a_{n_0}}{b_{n_0}} > 0.$$

Then

$$a_n \leq Cb_n, \quad \forall n \geq n_0.$$

Since

$$\sum_{n=1}^{\infty} b_n$$

converges, then

$$\sum_{n=n_0}^{\infty} b_n$$

also converges, and therefore

$$\sum_{n=n_0}^{\infty} Cb_n$$

converges.

From

$$0 < a_n \leq Cb_n, \quad \forall n \geq n_0,$$

by the comparison test,

$$\sum_{n=n_0}^{\infty} a_n$$

converges.

Finally,

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{n_0-1} a_n + \sum_{n=n_0}^{\infty} a_n.$$

The first sum is finite, and the second one converges. Hence

$$\sum_{n=1}^{\infty} a_n$$

converges.